



IoT ENABLED DC ENERGY METER

Our GSM-based meters deliver real-time DC power insights, empowering you to optimize consumption and boost efficiency, a leap forward in monitoring and managing overall consumption. This meter is more than just a meter; it's a gateway to a smarter, more sustainable future. Designed to embrace the power of precision with real time monitoring wide range of tenant loads for distributed energy monitoring.

4 CHANNEL

LOAD MONITORING

BI DIRECTIONAL

REAL TIME CLOCK

CLAMP-ON CT

FEATURES

- 4 channel DC energy totalizer.
- Measures common Bus Voltage and Current consumption of 4 channels.
- Comes with 4 split core CT's with a measuring range of 0 to 100A (more ranges available).
- Displays instantaneous I, V, KW and cumulative KWH.
- Accuracy: $\pm 1.0\%$ of FSD (5 % to 100 % Current range).
- 4G GSM Module built-in with Single SIM Interface.
- Password protected energy reset and DIN Rail Mountable.

TECHNICAL PARTICULARS

OFFERED SPECIFICATION

● Measurement Parameters	
● DC Voltage	1 Channel
● DC Current	4 Channels
● DC Supply	From Rectifier DC voltage (i.e., self-powered)
● Range of Measurement Parameters	
● DC Voltage	20 - 60 V DC
● DC current	1-100 A DC
● Resolution of Measurement Parameters	
● DC Voltage	0.1 V or better
● DC Current	0.2A or better
● Measurement Input Method	
● DC Voltage	2 pin Direct Input Connected to Rectifier Output
● DC Current	4 pin connectors for each channel
● Measurement Accuracy	
● DC Voltage	1% FSR @25 degree C or better
● DC Current	1% FSR @25 degree C or better, between 5 to 95% of FSR
● Temperature Coefficient	0.05% / degree C
● Derived Parameters	
● Power	kW
● Energy	kWh

● Range of Derived Parameters

● Power	0 - 65kW
● Energy	0 -100000 kWh

Displayed Resolution of Derived Parameters

● Power	0.1 kW
● Energy	0.1 kWh

● Accuracy of Derived Parameters

● Power	1% FSR @25 degree C or better
● Energy	1% FSR @25 degree C or better

● Displayed Parameters

● Cumulative Energy	All 4 channels
● Power	All 4 channels
● Ampere	All 4 channels
● Voltage	Common Rectifier Voltage

● Range of Editable Parameters

● Current rating	1- 100 A
● Slave ID	1 to 255
● Cumulative Energy	Reset all 4 channels (limit of software max value)
● Channel	Enable / Disable all 4 channels
● RTC	Real Time Clock Setting
● Password	Change password
● Front Panel Switches for Editing Input	2 - 4 Front Panel Mounted Membrane or Tactile

● Input Impedance

● Voltage	> 100 k ohms in Measured Circuit in Active State
● Current	> 10 k ohms Active State Per Channel

● Inbuilt Protection for Rugged Telecom Applications

● Voltage	1.2 times Rating & Reverse Polarity
● Current Channel	Non contact type measurement
● Editable Parameters	Accessible Only with Password
● Cumulative Energy (kWh)	Stored & Incremented in Memory

● Termination Connector on DCEM Back Plane

● Voltage Measurement Port & Supply Port	2 Pin Male / Female Pluggable for overall voltage
● Current measurement port	4 Pin Male / Female Pluggable for voltage per channel
● Communication Type	GSM 4G with Single SIM
● Contact material	Brass with Tin Plating
● Solder Pin Marking	Non Erasable Ink for all markings

● Internal Details of DCEM - Critical Components / Software Inputs

● Front Screen Display	16 x 4 Alphanumeric LCD with Backlight
● Memory	EEprom 64k
● Memory Life	5 Years or More Without Corruption
● RTC - Real Time Clock	High Accuracy with Button Battery Backup
● Update of Display Readings	Approx. 1 sec all 4 Channels Simultaneously
● Update of DCEM Memory	Every 0.1 kWh Increment
● Power Consumption	Approx. 4 Watts

● Environmental & General Specifications

● Operating Temperature	0 to 55 °C
● Storage Temperature	0 to 80 °C
● Operating Humidity	0 to 80 % RH Non-condensing
● Casing Material	Virgin Polycarbonate on All Sides, Heat Resistant Grade
● Size of DCEM	
● Colour	White

● Communication Interface

● Type	GSM 4G with Single SIM
● Protocol	HTTPS
● Settings -1	Meter Unique ID
● Settings - 2	Send Interval

CT SPECIFICATIONS

CT Type	Split Core
CT output	$\pm 2.5V @ \pm 100A$
Accuracy	$\pm 1.0\%$ of FSD (5 % to 95 % Current range)
Power Supply	+5V ($\pm 5\%$) Available from meter.
Linearity	$\pm 1\%FS$
Response time	<20ms
Inner Diameter	21mm
Outer Diameter	48mm
Thickness	32mm (including Connector)
Operating temperature range	0 ~ +85°C
Storage temperature range	0~+125°C



118,000+
Telecom Towers



700+
Solar Sites



15+ Global
MNO/Tower CO



26+
Countries



500,000+
Connected Assets



20,000+
Enterprise Users

DIN EN ISO 9001:2015

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MORE DETAILS**